Permissive Data Is Not the Bottleneck: Front-Loading Reasoning Data Enables Strong Post-Training

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Workshop: Transitioning from Pre-Training to Post-Training

NEURAL INFORMATION PROCESSING SYSTEMS 2026

Code, evals & checkpoints

Question & TL;DR

> Is **permissively licensed** pretraining data a bottleneck for post-training, or does the **composition** of the mixture matter more?

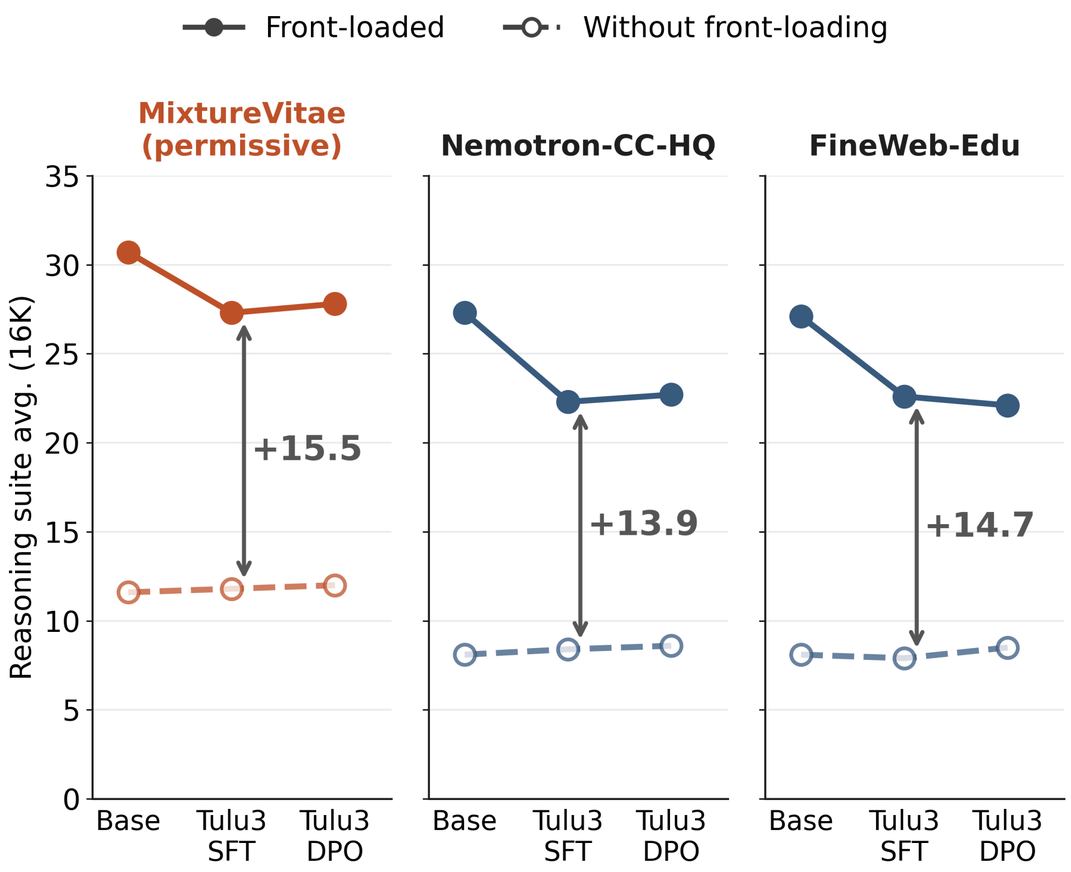
- **Front-loading** instruction & reasoning data into pretraining is the driver: **+10.3 to +19.2** pts of reasoning suite average in **all 18** cells.
- **Licensing is not the bottleneck for reasoning.** With front-loading, permissive **MixtureVitae** matches the non-permissive datasets at 4K and leads by **+4.7–5.0** pts at 16K.
- **Receptivity to reasoning SFT is laid down in pretraining.**
- **One gap remains:** general language understanding.

3 × 2 factorial design



- **1.7B models, 300B tokens**, identical open-sci-ref recipe and post-training: only the pretraining data changes.
- **Web dataset** (**MixtureVitae** permissive, Nemotron-CC-HQ, FineWeb-Edu) × **front-loading** (the same instruction & reasoning subset present or absent).
- **Evaluation:** 11-task reasoning suite (math, code, science QA, IFEval), 3 decoding seeds, paired bootstrap CIs; language understanding suite. Training data decontaminated (13-gram).

1 Front-loading drives the gain in every dataset



- **Tulu3 cannot substitute for front-loading**, although it contains math & code: the gap is largest at the base (~19 pts) and Phase 1 recovers only part of it.
- **Matched front-loading:** **MV** vs. Nemotron / FineWeb is **+0.5 / +0.4** at 4K (CIs include 0) and **+5.0 / +4.7** at 16K ($q \le 0.001$).
- **Broad gains** (16K, after SFT): **+23.0** code, **+15.6** math, **+10.4** instruction following.
- **Language understanding:** front-loading adds +5.0 (**MV**) and +5.2 (FineWeb), but **MV** still trails by **6.8–8.8** pts.

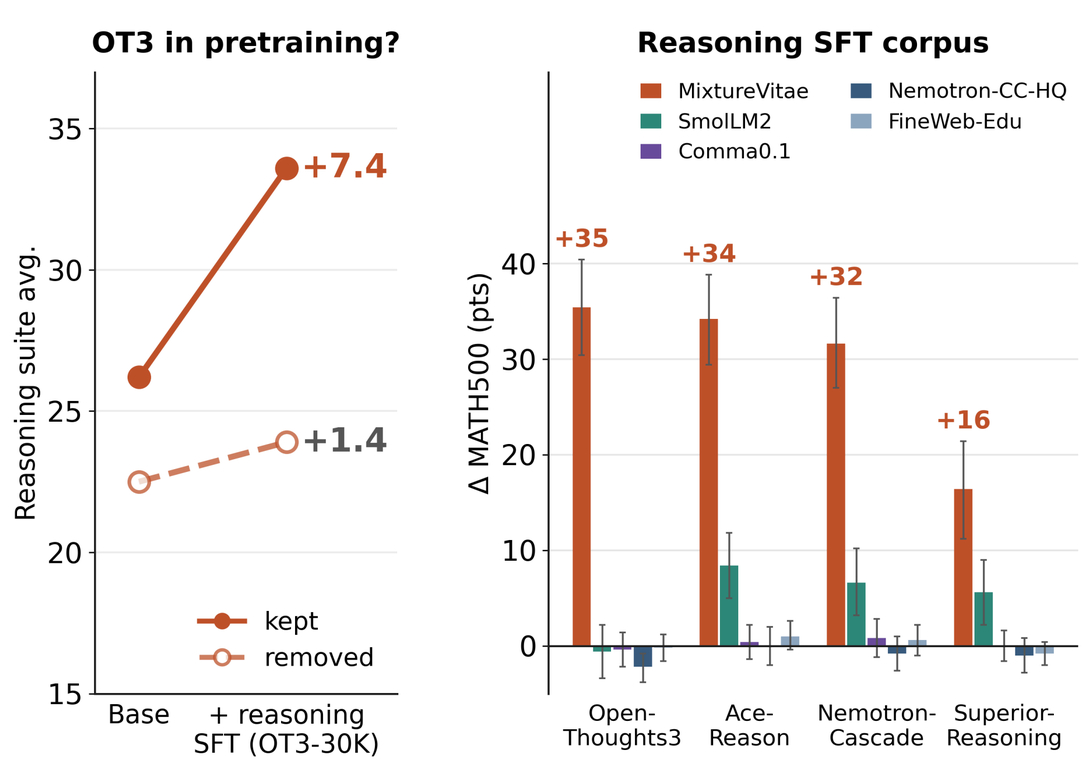
2 Across the model zoo

	Phase 1: Tulu3			Phase 2
Reasoning suite avg. (16K)	**Base**	**SFT**	**DPO**	**Δ reas. SFT**
Matched compute · 300B tokens				
MixtureVitae (front-loaded)	**30.7**	**27.3**	**27.8**	**+3.6***
Comma0.1 (perm., not enriched)	11.2	12.0	12.3	**−5.0**
Nemotron-CC-HQ	8.1	8.4	8.6	**−2.8**
FineWeb-Edu	8.1	7.9	8.5	**−2.4**
External references · 11–36T tokens				
SmolLM2 1.7B	17.1	18.4	19.1	**−4.3**
Qwen2.5 1.5B	10.1	24.0	25.4	−1.3
Qwen3 1.7B	30.9	31.6	31.7	**−19.3**

Δ reas. SFT: change under the identical OpenThoughts3-30K reasoning SFT from each model's Tulu3 SFT checkpoint (*MV from its base), scored with a longer sampled generation budget, so not comparable with the Phase 1 columns.

- **Comma0.1** is permissive but not enriched, and sits with the non-permissive references: **composition, not licensing, predicts the ranking**.
- Only the front-loaded permissive base **gains** from a small reasoning budget; every other base declines.

3 Receptivity is laid down in pretraining



- **Remove OpenThoughts3 from pretraining** and the same reasoning SFT reaches **23.9** instead of **33.6**: MATH500 **38.0** vs. **69.8**, and AIME24/25 fall from 11.7/10.0 to **0**.
- **Not re-exposure:** three corpora absent from pretraining give **MV** **+16 to +34** MATH500; no matched-compute reference improves with any corpus.
- **Cost:** instruction following drops under reasoning SFT (DPO first largely preserves it); language understanding does not.

Takeaways

- **Composition, not licensing,** decides post-trained reasoning: front-load instruction & reasoning data.
- **A permissive-first constraint is affordable** for math, code and reasoning; language understanding still lags.
- **Limitations:** 1.7B / 300B scale, one permissive pool, no RL stage.

Code, evals & checkpoints:
github.com/MixtureVitae/permissive-post-train